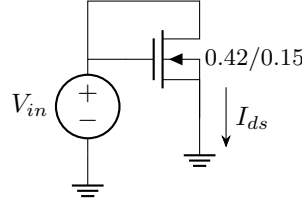


EE5311 Digital IC Design: Assignment 1
Transfer characteristics of transistors

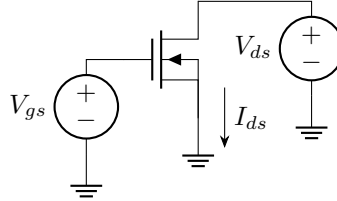
$$\begin{array}{llll}
 V_{Tn} = 0.7 \text{ V}, & \mu_n = 0.025 \text{ m}^2/\text{Vs}, & C_{oxn} = 8.34 \text{ fF}/\mu\text{m}^2, & v_{satn} = 8 \times 10^4 \text{ m/s}, & \lambda_n = 0.2/\text{V} \\
 |V_{Tp}| = 0.7 \text{ V}, & \mu_p = 0.009 \text{ m}^2/\text{Vs}, & C_{oxp} = 8.16 \text{ fF}/\mu\text{m}^2, & v_{satp} = 3 \times 10^4 \text{ m/s}, & \lambda_p = 0.2/\text{V}
 \end{array}$$

1. (a) Consider the nMOS transistor connected as shown below.



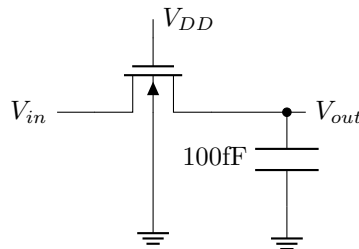
Is it in the saturation or linear region? Using Ngspice, obtain the transfer characteristics I_{ds} vs V_{gs} assuming $L = 0.15\mu\text{m}$ and $W = 0.42\mu\text{m}$. Plot the simulated model and analytical model in the same plot and find the mean percentage error.

- (b) Connect sources to the nMOS transistor as shown and obtain the output characteristics for $V_{gs} = 0.6\text{V}, 1\text{V}, 1.4\text{V}$ and 1.8V . Plot the analytical model and obtain the mean percentage error.



- (c) Simulate the I_{ds} vs V_{ds} for $V_{gs} = V_{DD}$ for transistors of different W/L keeping W/L ratio constant while increasing the sizes. Note how the velocity saturation effect decreases as overall length increases.
2. Repeat exercise 1 for pMOS transistor with same W and L .

3. (a) Connect a 100fF capacitor to the source terminal of an nMOS (with $L=0.15\mu\text{m}$ and $W=0.42\mu\text{m}$) as shown below. Provide a step input to V_{in} at $t = 0$ going from 0 to $V_{DD} = 1.8\text{V}$ with a rise time of 10ps . Use the PULSE waveform to do this (refer to Tutorial 0). Assuming that the capacitor is completely discharged at $t = 0$, find V_{out} at $t = 20\text{ns}$ and 100ns . How do these values compare against the V_{out} at $t = \infty$ calculated using the analytical model?
(Hint: Use .ic SPICE command to set initial conditions. e.g. .ic V(Vout)=0)



- (b) Repeat part 3(a) with a pMOS ($L=0.15\mu\text{m}$ and $W=0.42\mu\text{m}$) whose gate is grounded and body is connected to V_{DD} and the capacitor is connected to its drain.
- (c) Repeat part 3(a) for falling waveform at V_{in} going from V_{DD} to 0 with a fall time of 10ps . Assume that the capacitor is charged to V_{DD} at $t = 0$.
- (d) Repeat part 3(c) with a pMOS ($L=0.15\mu\text{m}$ and $W=0.42\mu\text{m}$) whose gate is grounded and body is connected to V_{DD} and the capacitor is connected to its source.